Shekun Wang

Uncertainty Quantification · Surrogate Modeling
Scientific Machine Learning

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Research Interests

Scientific machine learning; uncertainty quantification; high-dimensional, limited-data surrogate modeling; full-field surrogate and operator modeling; robust design optimization.

Education

Northwestern Polytechnical University Sep 2024 – Apr 2027
M.Eng. in Aerospace Science and Technology Xi'an, China

- GPA: 3.918/4.1; weighted average: 92/100.
- Supervisor: Huan Zhao.
- Thesis topic: *Sparse Polynomial Chaos-Based Multidisciplinary Robust Aerodynamic and Sonic-Boom Design*.

Northwestern Polytechnical University Sep 2020 – Jun 2024
B.Eng. in Aircraft Design and Engineering Xi'an, China

- GPA: 3.852/4.1; rank: 9/178 (top 5.1%).
- Honors: Outstanding Graduate.
- Thesis: *Sparse Polynomial Chaos-Based Low-Drag, Low-Boom Robust Design Optimization of Aircraft Shape*.

Research Experience

Graduate Researcher – Uncertainty Quantification and Surrogate Modeling Sep 2023 – present
School of Aeronautics, Northwestern Polytechnical University; Advisor: Huan Zhao

- Develop structure-exploiting, data-efficient surrogate models for UQ of computationally expensive systems, with emphasis on high-dimensional, limited-data, and field-valued problems.
- Exploit intrinsic response structures, including sparsity, low dimensionality, gradient compressibility, and spatial locality, to improve surrogate accuracy, stability, and scalability.
- Carry methodological ideas from mathematical formulation and theoretical analysis to modular MATLAB implementations for surrogate construction and uncertainty propagation.
- Evaluate methods on analytical benchmarks, stochastic PDEs, and transonic CFD, and develop Python/HPC workflows linking surrogate-based UQ to aerodynamic and sonic-boom robust design optimization.

Journal Publications and Manuscripts

Published Articles

- **Shekun Wang**, Huan Zhao, and Keyao Gan. "Sparse Arbitrary Polynomial Chaos for High-Dimensional Problems Based on Moment-Constrained Projection Dimensionality Reduction." *Reliability Engineering & System Safety*, 279, 113773, 2027. doi:10.1016/j.ress.2026.113773. IF 13.7 JCR Q1

  Develops MCP-aPC for high-dimensional, limited-sample UQ with nonstandard or partially characterized inputs, preserving low-order moments and regularizing second-order structure under an orthogonality constraint. LOO-selected models improve accuracy and robustness on analytical problems up to 300 dimensions, a stochastic heat-transfer PDE, and M6-wing aerodynamics.

- **Shekun Wang** and Huan Zhao. "Gradient-Residual Aware Subspace-Enhanced Arbitrary Polynomial Chaos-Kriging for High-Dimensional Surrogate Modeling." *Engineering with Computers*. IF 4.1 JCR Q1

  Develops a GRAS-based low-rank extension of gradient-enhanced aPCK for limited-data, high-dimensional modeling, replacing the full set of coordinate-wise derivative channels with projected directions that retain dominant gradient activity and trend-unexplained content. The formulation reduces the augmented system from $N(d+1)$ to $N(r+1)$ while preserving the GE-aPCK predictor and projected-covariance validity; analytical, PDE, and aerodynamic benchmarks approach full GE-aPCK accuracy at substantially lower training cost.

- **Shekun Wang**, Huan Zhao, Zhengang Zhang, and Weifeng Zhang. “An adaptive weighted lasso regression method for sparse polynomial chaos expansion: Application to multi-source uncertainty quantification.” *Reliability Engineering & System Safety*, 275, 112832, 2026. doi:10.1016/j.ress.2026.112832. IF 13.7 JCR Q1

  Applies adaptive weighted Lasso regression to sparse PCE for high-dimensional, data-scarce UQ, using preliminary coefficient estimates to assign term-specific ℓ_1 penalties that stabilize basis selection and reduce shrinkage of influential terms. Extends the analysis to establish selection consistency, asymptotically unbiased estimation, and an approximation–statistical convergence characterization; analytical benchmarks, transonic aerodynamic analysis, and sonic-boom propagation studies demonstrate improved robustness and sample efficiency.

- Zhengang Zhang, Huan Zhao, **Shekun Wang**, Kaiyuan Fan, and Zhenghong Gao. “Multipoint Inverse Design Method for an Integrated Tail-Propulsive Propeller in Canard Rotor-Wing UAV.” *IEEE Transactions on Aerospace and Electronic Systems*, 62, 10427–10445, 2026. doi:10.1109/TAES.2026.3689077. IF 7.0 JCR Q1

  Conducted the CFD simulations for aerodynamic performance evaluation and validation of the propeller designs across static-thrust and high-advance-ratio operating conditions.

- Huan Zhao, **Shekun Wang**, Zhenghong Gao, and Jiangtao Huang. “Research progress on aerodynamic robust design optimization methods for air vehicles.” *Acta Aerodynamica Sinica*, 42(9), 37–69, 2024. doi:10.7638/kqdlxxb-2023.0234.

  Authored the sections on uncertainty quantification, surrogate modeling, and robust aerodynamic and sonic-boom design, synthesizing representative methods, applications, and open challenges.

Manuscripts

- UNDER REVIEW **Shekun Wang** and Huan Zhao. “A Sparse Partition-of-Unity Arbitrary Polynomial Chaos Operator for Full-Field Surrogate Modeling.” *Journal of Computational Physics*. IF 3.9 JCR Q1

  Develops S-PoU-aPCO for continuous full-field UQ from scarce snapshots, coupling a data-driven stochastic basis with overlapping local physical spaces and adaptive weighted ℓ_1 regression of physical–stochastic couplings. The formulation shares information across the field while retaining continuous physical evaluation, compatible boundary constraints, and coefficient-based analytical moments; stochastic-transport and Darcy-flow PDEs and an airfoil CFD benchmark show improved accuracy when dense identification is underdetermined.

Conference Papers and Oral Presentations

- **Shekun Wang** and Huan Zhao. “Sparse Polynomial Chaos-Based Robust Optimization Design of Aerodynamic and Sonic Boom.” In *Proceedings of the 35th Congress of the International Council of the Aeronautical Sciences (ICAS 2026)*, Sydney, Australia, 2026. Oral presentation. EI SCOPUS

  Develops a non-intrusive sparse PCE framework for limited-data aerodynamic and sonic-boom UQ and robust design, demonstrated on a supersonic wing–body under coupled uncertainties in Mach number, angle of attack, Reynolds number, altitude, and humidity.

- **Shekun Wang**, Huan Zhao, Keyao Gan, Zhiyuan Gong, and Yujie Gan. “Sparse Polynomial Chaos-Based Sonic Boom Uncertainty Quantification and Aerodynamic/Sonic Boom Robust Design Optimization under Multi-Parameter Uncertainties.” In *Proceedings of the 34th Congress of the International Council of the Aeronautical Sciences (ICAS 2024)*, Florence, Italy, 2024. Oral presentation. EI SCOPUS

  Validates AFBS-based sparse PCE for far-field sonic-boom UQ against OMP, OLS, and Monte Carlo, with augmented Burgers propagation under six uncertainties (Mach number, altitude, temperature, humidity, wind direction, and wind speed). A separate robust-design study optimizes 30 FFD geometry variables by Kriging-assisted particle swarm optimization and evaluates aerodynamic and PLdB statistics under Mach-number and angle-of-attack uncertainty.

Selected Research Projects

National Natural Science Foundation of China (NSFC)

Bilevel-Nested Gradient-Enhanced Dimensionality-Reduction Surrogate Modeling for High-Dimensional Global Robust Aerodynamic Design Optimization

Grant No. 52572420 • 2026–2029

- Within the project's broader objective of high-dimensional robust aerodynamic design, contribute surrogate-modeling and UQ methods for settings characterized by expensive high-fidelity CFD, high-dimensional inputs, coupled uncertainties, and limited training data.
- Investigate structure-exploiting strategies based on sparse polynomial representations, response-informed dimension reduction, compressible gradients, and spatial locality to improve data efficiency and scalability in surrogate construction and uncertainty propagation.
- This work has produced adaptive weighted Lasso PCE, MCP-aPC, GRAS-based low-rank GE-aPCK, and S-PoU-aPCO for sparse recovery, dimension reduction, scalable gradient use, and data-efficient continuous full-field surrogate modeling through adaptive sparsity, respectively.

High-Dimensional Global Aerodynamic Optimization via a Novel Multi-Fidelity Polynomial Chaos–Kriging Model

Grant No. 12102489 • 2022–2024

- Within a broader project on high-dimensional global aerodynamic optimization, contributed SKPCA–PCK methodology for nonlinear latent-space construction and PCK calibration, using nested optimization to improve high-dimensional, limited-sample surrogate modeling.

Research Software

Surrogate Modeling and Uncertainty Quantification Framework MATLAB

- Developed a modular framework for data-efficient surrogate modeling and UQ, integrating arbitrary polynomial chaos (aPC), gradient-enhanced polynomial chaos–Kriging (GE-aPC-K), sparse regression, dimensionality reduction, model selection, scalar and full-field prediction, uncertainty propagation, and sensitivity analysis.

CFD-Based Robust Design Optimization Framework Python

- Developed an automated surrogate-assisted robust optimization workflow integrating SU2 aerodynamic evaluation, near-to-far-field sonic-boom propagation via the augmented Burgers equation, subsequent perceived-level (PLdB) evaluation, Kriging modeling, aPC-based UQ, and iterative high-fidelity refinement. Extended the open-source packages pyGeo with three-dimensional wing CST parameterization and IDWarp with native SU2 mesh I/O.

ParaView SU2 Mesh Reader Python

- Developed a public ParaView plugin for direct visualization and inspection of SU2 volume and boundary meshes.

Scholarships

National Scholarship for Graduate Students	2026, 2024
Wu Yajun Scholarship, Third Prize	2025, 2022
Northwestern Polytechnical University First-Class Scholarship	2026, 2025, 2024, 2022

Technical Skills

Programming	MATLAB, Python
Methods	Polynomial chaos expansion, Kriging, sparse regression, dimensionality reduction, gradient-enhanced surrogate modeling, uncertainty quantification, sensitivity analysis, robust optimization
CFD & Engineering Tools	SU2, Pointwise, IDWarp, Tecplot, ParaView
Computing Environment	Linux, HPC clusters